

# Expense Tracker with Budget Alert – Planning Document

## 1. Introduction

This project is a command-line based Expense Tracker designed to help users manage their daily expenses and monitor their monthly budget. It allows users to record expenses, categorize them, and analyze their spending habits. The system also provides alerts when the total expenses exceed the predefined budget.

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## 2. Objective

The main objective of this project is to develop a simple and efficient expense tracking system that:

- Records daily expenses
- Categorizes expenses (food, travel, etc.)
- Calculates total and category-wise expenses
- Allows setting a monthly budget
- Alerts the user when the budget is exceeded

Additionally, the project demonstrates DevOps practices using Git for version control and Docker for containerization.

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## 3. System Design

### 3.1 Data Structure

- Expenses are stored in a list of dictionaries
- Each expense contains:
  - Amount
  - Category
  - Date

### 3.2 Workflow

1. User inputs expense details
  2. Data is stored in memory
  3. User can view all expenses
  4. System calculates totals
  5. Budget is compared with expenses
  6. Alert is generated if budget exceeds
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## 4. Features

- Add Expense (with category and date)
  - View All Expenses
  - Show Category-wise Total
  - Set Monthly Budget
  - Budget Alert System
  - Date-wise Expense Filtering
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## **5. Tools and Technologies**

- Programming Language: Python
  - Version Control: Git
  - Containerization: Docker
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## **6. Conclusion**

The Expense Tracker project successfully demonstrates the implementation of basic programming logic along with DevOps concepts. It provides a practical solution for managing expenses and can be extended further with additional features and improvements.